Creation and Evaluation of Human Models with Varied Walking Ability from Motion Capture for Assistive Device Development

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***Abstract*—As the world ages, rehabilitation and assistive devices will play a key role in improving mobility. However, designing controllers for these devices presents several challenges, from varying degrees of impairment to unique adaptation strategies of users. To use computer simulation to address these challenges, simulating human motions is required. Recently, deep reinforcement learning (DRL) has been successfully applied to generate walking motions whose goal is to produce a stable human walking policy. However, from a rehabilitation perspective, it is more important to match the walking policy's ability to that of an impaired person with reduced ability. In this paper, we present the first attempt to investigate the correlation between DRL training parameters with the ability of the generated human walking policy to recover from perturbation. We show that the control policies can produce gait patterns resembling those of humans without perturbation and that varying perturbation parameters during training can create variation in the recovery ability of the human model. We also demonstrate that the control policy can produce similar behaviours when subjected to forces that users may experience while using a balance assistive device.**

I. INTRODUCTION

The World Health Organisation (WHO) predicts that by 2030, one in six people in the world will be aged 60 years or over [1]. From a physical ability standpoint, this rise will lead to more gait-related impairments. These impairments lead to a significant decrease in quality of life for older people as they highly value their mobility [2].

In recent years, there has been a rise in robotic assistive walking devices [3, 4] and exoskeletons [5] that help improve the gait stability of the users [6, 7]. However, the development of these devices faces several challenges [8]. Firstly, designing a controller for multiple types of gait impairment is challenging due to the variation of impairment among users, which is influenced by many factors. Secondly, users develop unique adaptation strategies with the device that the controller may not anticipate. Lastly, testing of such devices is often limited or impractical due to the vulnerability of the target users, leading to limited data.

Computer simulation can address these developmental challenges by providing insights into the interaction between the device and the user [9, 10, 11]. One of the key elements here is the simulation of human motions. Recently, simulating human motions has been increasingly driven by a control policy (CP) learned through deep reinforcement learning (DRL) in physics-based environments [12, 13]. DRL methods have many variations. These include constructing a reward function based on reference motion [12, 13], curriculum learning [14], combining pose-dependent joint limits with muscle activation dynamics [15] or learning a joint-torque controller from muscle activation data [16].

With respect to walking motions, during the DRL training process, the learned control policy will learn some ability to recover from perturbation [13, 17, 18]. Additionally, it has been reported that randomly pushing the human during training simulations can result in increased robustness [19, 20]. This is expected as part of their exploration vs exploitation trade-off in DRL as the learning converges to producing realistic human motion. However, while these learned CPs can handle some small disturbances, they often fail to recover when subjected to perturbations that are present during locomotion [21]. Park *et al.* [22] showed that DRL policies produce push-recovery stability that is as robust as human walking with their lateral disturbance investigation. From a rehabilitation perspective, users of assistive and rehabilitation devices often have reduced physical ability. However, no research has been done in trying to investigate the correlation of DRL training parameters with the ability of the walking policy. More specifically, understanding what it takes to match the walking policy's ability to that of an impaired person with reduced ability.

In this paper, we present the first attempt to correlate DRL training parameters with the ability of the generated human walking policy, specifically the ability to recover from perturbation. We first create a 3D personalised human model based on real motion capture data using OpenSim [23] and convert it to MuJoCo [24] for simulation in a physics-based environment. Subsequently, we train a human walking policy that is conditioned on the motion capture data of the subject. Our human walking policy is inspired by Peng *et al.* [17] using adversarial motion priors to produce CPs that generate realistic human motions using unstructured motion clips with a simplified reward function that requires minimal tuning. In addition to imitating the human's gait, we also show ways to vary this developed control policy by varying perturbation parameters during training, thereby creating variation in the recovery ability of the human model.

Our assessment shows that the CPs can produce gait patterns resembling that of humans without perturbation. Additionally, we show the differences in the ability of the various trained CPs quantitatively when subjected to various levels of perturbation during training. Lastly, we show that the control policy can produce similar behaviours when subjected to forces that the user may experience when using an overground balance assistive robot.

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Overall, our study contributes to the understanding of how introducing perturbation can influence the recovery ability of walking policies, shedding more light on the creation of personalised human models with varied abilities that can be used for the development of assistive devices.

II. METHOD

A. 3D Personalised Human Model

We define a full-body skeletal model that contains 27 degrees of freedom (DoFs), is physically simulated and actuated by joint torques in the open-source physics engine MuJoCo [24]. It has a six DoFs un-actuated root joints defined at the pelvis, five ball joints at the (hip, shoulder, and lumbar), and six revolute joints (knee, ankle, elbow). A 3D personalised human model is created using the motion capture data of a subject captured using the experiment protocol [25] defined by the Rehabilitation Research Institute of Singapore (RRIS). The static motion capture data captures the body anthropometric, and the dynamic motion capture data captures the baseline ability of the subject.

The human model is scaled from a generic well-documented OpenSim model [26] using the in-built scaling tool given the RRIS marker set placed on the subject during motion capture. We apply the dynamic motion capture data to this scaled model and utilise the inverse kinematics function to extract the various gait parameters. In particular, joint angles per frame, which will be used later as part of the training data when creating the human walking policy. We utilise open-source converters [27] to convert the OpenSim model to MuJoCo as seen in Figure 1. In this paper, we do not perform any muscle-related analysis; hence, we simplify the musculoskeletal model down to a rigid body skeletal model to enjoy performance benefits during DRL training of the human walking policy. To complete the model definition, we define a torque actuator on each joint on the skeletal model.

Fig. 1. **Left:** Converted Musculoskeletal Model in MuJoCo **Right:** Simplified Skeletal Model in MuJoCo derived from Musculoskeletal Model. Each blue arrow in both images represents joints axis of rotation.

B. Human Walking Policy

We use a model-free DRL approach to develop a human control policy $\pi_h(\mathbf{a}_h|\mathbf{s}_h)$, where the policy outputs an action $\mathbf{a}$ given the state $\mathbf{s}$ observed in the environment. To produce naturalistic walking, we provide a single reference motion of the subject doing 10m walking. We utilise the network architecture defined by Peng *et al.* [17] and closely follow their definition of states and discriminator observations.

Different from Peng *et al.* [17], we simulate all environments in MuJoCo [24], and our implementation utilises the Python API provided by OpenAI with MuJoCo-Py. The policy is queried at 30Hz with a simulation frequency of 500Hz. The policy outputs an action that specifies a target position for PD controllers positioned at the character's joints. The PD controller will convert this target action to output torque to actuate the joints [28]. The action space has 21 dimensions after excluding the un-actuated six DoF root joints. To provide the human agent with more information during training, we also utilise the reference state initialisation technique [13], where the initial pose and velocity of the agent are randomly sampled from all possible poses in the reference motion during initialisation of the simulation.

In addition to the various early termination criteria defined by Peng *et al.* in [13, 17], we terminate early after a specific pelvis height threshold. We define the character as fallen when the pelvis height has fallen to 70% of the initial pelvis height. This allows the training of the policy to progress faster, especially in the early stages of the exploration.

C. Varying Human Walking Policy's Recovery from Perturbation Ability

A common way to increase the robustness of a CP developed through DRL is to increase the diversity and samples experienced during training [21, 22]. We introduce perturbation during CP training, exposing the model to a diverse range of disturbances to expand its experience buffer. This promotes learning of adaptive CPs that can generalise strategies to different perturbation scenarios. Our perturbation parameters are selected for their real-world implications. These include the timing, position, duration, magnitude, and direction of perturbation. Timing simulates disturbances at different moments, while positions represent the impact on different body parts. Duration affects how long the perturbation is applied, such as a constant perturbation or instantaneous perturbation. Magnitude and direction mimic the forces and directions encountered. With these parameters, we aim to recreate different disturbances that a real person may face, expanding the agent's ability to recover and adapt.

Our baseline is a CP that does not experience any perturbation during training. We also train two CPs conditioned on experiencing random perturbations during training. For each simulation instance, we apply a constant perturbation magnitude randomly sampled from [0N, 50N] and [0N, 200N] for the two policies, respectively. The direction of perturbation is fixed to the plane parallel to the ground, and for each instance, the direction is uniformly sampled between $[-\pi, \pi]$ and is fixed throughout that instance. The perturbation is applied 0.1 seconds after the human model is loaded into the simulation to provide time to take initial action and is applied throughout the simulation instance until any of the termination criteria is met. We exert the perturbation on the CoM of the human model as research has shown that the position of CoM is critical to the recovery ability of the person during walking [29, 30]. Table I shows the list of CPs trained and variation in perturbation parameters. All CPs are trained on the same single reference motion clip of a subject walking in a straight line. By randomising the above parameters, we ensure that there is appropriate diversity to populate the experience replay.

TABLE I
NAME OF EACH CONTROL POLICY WITH THE VARIOUS PERTURBATION PARAMETERS

Name	Force Range	Start Time	Duration
CP-BASE	0N	Not Applicable	Not Applicable
CP-A	[0N, 50N]	0.1s onwards	Till Termination
CP-B	[0N, 200N]	0.1s onwards	Till Termination

III. EXPERIMENTS AND RESULTS

The goal of the experiments is to determine if the trained human CPS not only produce gait patterns resembling that of the human but also whether its ability to recover from perturbation can be tuned by leveraging on the exploration vs exploitation properties of DRL. Our experiment framework provides a deeper understanding of the correlation between perturbation parameters during training and the recovery ability of the trained model. To obtain the gait information for the reference motion, we utilise OpenSim [23], following their recommendations for inverse kinematics [31], and inverse dynamics [32]. In our experiments, the human model is from a male subject's motion capture data (1.68 meters, 59.9 kg). Participation in the motion capture session was voluntary and informed consent was obtained. IRB-2018-04-014 governs the motion capture, and access to the data to create a non-identifiable 3D human model is governed by IRB-2022-713, approved by Nanyang Technological University's Institutional Review Board. For better visualisation of results, we would like to direct you to our video[1].

A. *Gait Evaluation of Control Policies with No Perturbation*

First, we compare the gait patterns generated against the reference motion to validate if the walking policy imitated the gait pattern accurately. For a representative gait pattern of each control policy, we collect 3 gait cycles for each simulation instance with 10 simulation instances per control policy, for a total of 30 gait cycles per control policy. Figure 2 shows generated gait by the various walking policies and the reference gait along the sagittal plane plotted from right heel strike to right heel strike and Table II shows the average stride length of each control policy.

From Figure 2, we see that CP-BASE, CP-A and CP-B produced gait patterns similar to the reference motion in shape with some minor differences in the range of joint angles. We believe that the slight mismatch of the range is due to the CPs prioritising stability and balance over achieving exact replication, which is a crucial aspect of CP training in the presence of disturbances. By favouring stability, the CPs might converge to a more conservative range of motion

[1]https://youtu.be/FYTPnA0w9SU

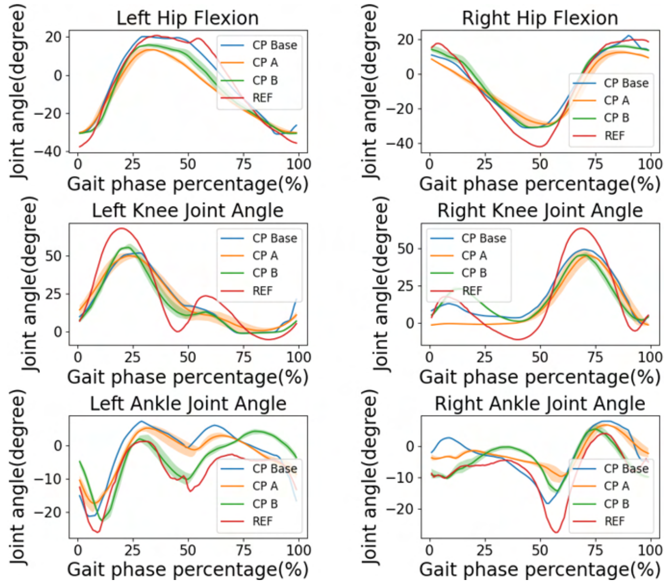

Fig. 2. Comparison of mean and standard deviation of lower limb joint angles of CP-BASE, CP-A and CP-B against reference motion along the sagittal plane.

compared to the more extensive range exhibited by the reference gait. We also note that the stride length of the CPs decreases, as seen in Table II as the perturbation magnitude increases. Similarly, we believe that the agent prioritised stability over achieving the goal of exact imitation, as stride length is only implicitly enforced during training through joint angles and not explicitly enforced during training. While there is a slight mismatch against the reference motion, the three CP's gait kinematics are close to one another, allowing us to compare their trained recovery ability.

TABLE II
COMPARISON OF STRIDE LENGTH BETWEEN REFERENCE MOTION, CP-BASE, CP-A AND CP-B.

Name	Mean(m)	Min(m)	Max(m)
Reference	1.56	Not Available	Not Available
CP-BASE	1.33	1.27	1.38
CP-A	1.21	1.17	1.35
CP-B	1.10	1.08	1.12

We also observe that CP-B developed a consistent tip-toe gait best seen in the supplementary video. We posit that the agent learned the tip-toe gait as an attempt to learn walking while being continuously perturbed. When a video comparison of the intermediate training progress is made against CP-BASE or CP-A, we observe that CP-BASE and CP-A agents can take a few steps of "normal" gait, which guides the agent to perform these actions more as these "normal" gait results have high imitation rewards. Conversely, we observe that the early onset of the large perturbation for CP-B during each simulation instance resulted in the agent not experiencing what a "normal" gait is, and a tip-toe gait is a form of exploitation taken by the RL agent to maximise the imitation reward and maintain balance despite the perturbation.

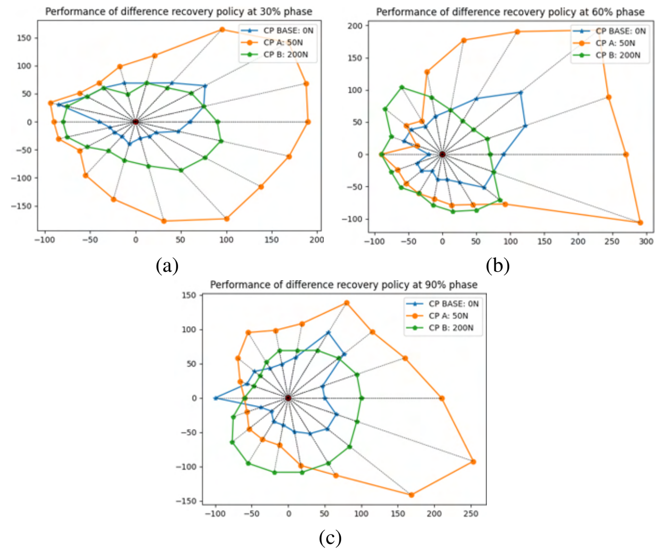

Fig. 3. Range of external perturbation required for each control policy before it fails to recover across a range of $[-\pi, \pi]$ at (a) heel rise (30% of the gait cycle), (b) toe off (60% of the gait cycle) (c) tibia vertical (90% of the gait cycle). Positive X is the forward direction of model. X and Y axes represent magnitude of force (N).

B. Gait Evaluation of Control Policies with Varying Perturbation

To test the recovery ability against perturbation, we perturb the trained walking policies across a range of $[-\pi, \pi]$ for 0.5 seconds to determine the force required such that the control policy fails to recover. We increment the magnitude of the force by 10N and the direction of perturbation by 20 degrees per simulation instance. Each combination of force and direction is repeated 5 times per control policy to get a representative average. For a combination of force and direction to be accepted, it must have a minimum of 80% pass rate. We refer to the definition of gait cycle with 0% at initial contact of the right foot [33] and perturb the model at the heel rise, toe-off and tibia vertical, which corresponds to approximately 30%, 60% and 90% of the gait cycle.

After perturbation, the policies will struggle for a short period of time before it either fails to recover or successfully recovers. For a successful recovery, the generated gait pattern by the walking policy must match within one standard deviation the representative gait pattern of that specific control policy. Our failure case occurs when the human model's pelvis falls below 70% of the original pelvis height during simulation initialisation. To provide ample time to recover, we also allow 3 seconds of simulation post-perturbation and if the model is still unable to reach the success criteria, we consider the model to have failed to recover. We showcase the varying ability to resist perturbation of the model by expanding on the stability region definition defined by Kumar *et. al* [21]. Figure 3 shows the stability region of each control policy in various directions and the maximum force it resisted in that direction before failing to recover.

From Figure 3, we observe that the CP-BASE and CP-A are much more stable when perturbed forward in the positive x-direction and leftward in the positive y-direction rather than backward or rightward, and this pattern of stability is consistent across the various time when the perturbation occurs. This is expected as when the model is initialized per simulation instance, the initial pose and velocity are applied to the joints that move the model towards a forward walking motion. Furthermore, the reference motion used to train the CPs is where the subject walks straight in a forward direction. This combination makes it unlikely that the model will explore walking backwards and hence, limit the capability of the CPs to resist pulling forces. However, the stability region for CP-B is less biased towards a specific direction than CP-BASE and CP-A. We posit that due to the tip-toe gait of CP-B, the model is more reactive to perturbation in all directions as compared to CP-BASE and CP-A's gait pattern, thereby explaining the more circular shape of the stability region of CP-B. In addition, we also note that CP-A's stability region is generally much larger than CP-BASE and CP-B. The larger size of CP-A relative to CP-BASE is an expected result as it has been proven that applying perturbation during training can increase the stability of the model [21, 22]. The larger stability of CP-A as compared to CP-B is surprising and unexpected. We posit that the tip-toe gait of CP-B leads to an earlier onset of instability during experimental conditions where we apply the perturbation only temporarily and not throughout the simulation instance. This "on-off" application of perturbation results in a change of perturbation that is too sudden for CP-B, leading to earlier onset of instability.

C. Adaptation to External Perturbation

Existing overground balance assistive robots such as MRBA [4] have well-documented literature highlighting that user experiences either forward push or rearwards pull due to system inertia and response time when attached to these devices as they walk forward, which is especially noticeable when the user changes their gait speed. To showcase that the human model has a similar reaction when experiencing either forward push or rearwards pull force, we run CP-A for 10 instances with a force with various constant forces based on a percentage of body weight for 0.2 seconds at 30% of the gait cycle starting from the right heel strike. This is shown in Table III together with their corresponding stride length.

TABLE III
COMPARISON OF STRIDE LENGTH FOR CP-A AFTER PERTURBATION FORCE BASED ON PRECENTAGE OF BODY WEIGHT.

Direction	% Weight(N)	Mean(m)	Min(m)	Max(m)
Forward	40	1.32	1.31	1.33
Forward	20	1.30	1.23	1.33
None	0	1.17	1.15	1.22
Rearward	20	1.10	0.94	1.07
Rearward	40	0.77	0.67	0.83

As the perturbation force is applied on the sagittal plane, we plot the lower limb gait kinematics on the sagittal plane where the effects is the most prominent in Figure 4. We note that there is a distinct phase shift post-perturbation and a slight change in magnitude on the joint angles and stride

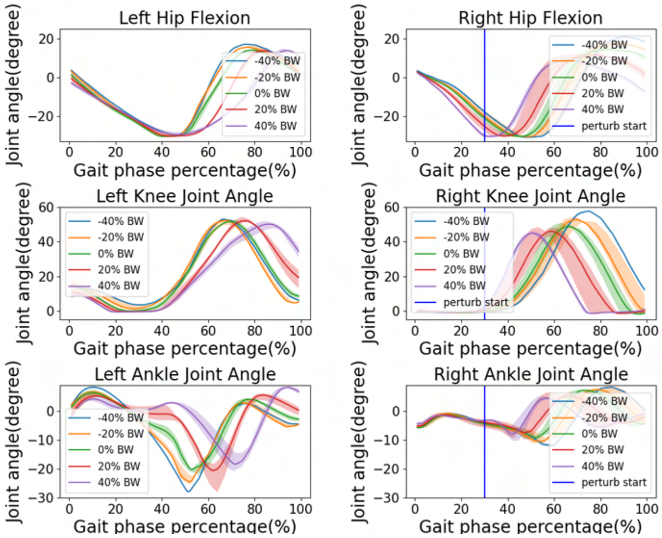

Fig. 4. Comparison of lower limb joint angles of CP-A with and without perturbation in the sagittal plane normalised to each foot's heel strike to heel strike. Negative % indicates the model is pushed forward. Perturbation applied at 30% of gait phase from right heel strike.

length proportional to the magnitude and direction of the perturbation. These changes to the gait are well reported in existing literature such as [4], and our model's ability to replicate similar reactions provide promise for future usage as part of assistive device development. For clearer visualisation of the changes, a video clip of CP-A experiencing rearwards perturbation and its reaction is added.

IV. CONCLUSIONS AND FUTURE WORK

We have proposed a method to create personalised 3D human models from motion capture data with varied walking ability. This was achieved by varying perturbation parameters during deep reinforcement learning of the human walking policy. We have shown that the CPs produce gait patterns resembling those of humans without perturbation. We also showed that the generated policies have different recovery ability when subjected to external perturbation. Lastly, we demonstrated that the control policy produces changes in gait patterns when subjected to forces that users may experience while using a balance assistive device.

In this work, we investigate the effect of varying perturbation force, duration, and frequency on the end ability of a DRL-trained human walking policy. We will continue investigating how other parameters such as the position of perturbation, incorporation of tasks, varying terrain, and more reference motions, will affect the end ability of the user. In particular, we are keen to investigate the reaction of the model in balancing the requirements given a task and external disturbances, such as those coming from rehabilitation and assistive devices, providing an unprecedented insight into the potential adaptations and behaviors the human might take. Additionally, we will apply our research to personalised musculoskeletal models to better understand the recovery ability on the muscle level. This will potentially enhance the model's recovery ability, allow us to control the model better, and replicate various musculoskeletal disorders, making the model more personalised for assistive device development.

We believe this is just the first step in providing greater granular control over the end ability of the human model. With a better understanding, we believe that we can create an accurate human model that provides a good representation of impaired individuals with reduced physical abilities. This allows for greater use of human-in-the-loop rehabilitation and assistive robotics simulation and provides an essential prototyping and testing guinea pig for developing rehabilitation and assistive devices.

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